

### EXERCISE 18

Structure of 'restaurants' collection:

```
{
  "address": {
    "building": "1007",
    "coord": [ -73.856077, 40.848447 ],
    "street": "Morris Park Ave",
    "zipcode": "10462"
  },
  "borough": "Bronx",
  "cuisine": "Bakery",
  "grades": [
    { "date": { "$date": "1393804800000" }, "grade": "A", "score": 2 },
    { "date": { "$date": "1378857600000" }, "grade": "A", "score": 6 },
    { "date": { "$date": "1358985600000" }, "grade": "A", "score": 10 },
    { "date": { "$date": "1322006400000" }, "grade": "A", "score": 9 },
    { "date": { "$date": "1299715200000" }, "grade": "B", "score": 14 }
  ],
  "name": "Morris Park Bake Shop",
  "restaurant_id": "30075445"
}
```

1. Write a MongoDB query to find the restaurant Id, name, borough and cuisine for those restaurants which prepared dish except 'American' and 'Chinees' or restaurant's name begins with letter 'Wil'.

db.restaurants.find( { \$or : [ { name : { \$regex : /Wil/ } }, { cuisine : { \$in : [ "American", "Chinees" ] } } ] }, { restaurant\_id : 1, name : 1, borough : 1, cuisine : 1 } );

2. Write a MongoDB query to find the restaurant Id, name, and grades for those restaurants which achieved a grade of "A" and scored 11 on an ISODate "2014-08-11T00:00:00Z" among many of survey dates..

db.restaurants.find( { \$and : [ { \$elemMatch : { grade : "A", score : 11, date : { \$date : "2014-08-11T00:00:00Z" } } } ] }, { restaurant\_id : 1, name : 1, grades : 1 } );

3. Write a MongoDB query to find the restaurant Id, name and grades for those restaurants where the 2nd element of grades array contains a grade of "A" and score 9 on an ISODate "2014-08-11T00:00:00Z".

db.restaurants.find( { "grades.1.grade" : "A", "grades.1.score" : 9, "grades.1.date" : { \$date : "2014-08-11T00:00:00Z" } } );

4. Write a MongoDB query to find the restaurant Id, name, address and geographical location for those restaurants where 2nd element of coord array contains a value

db.restaurants.find( { "address.coord.1" : { \$lt : -73.704100 } }, { restaurant\_id : 1, name : 1, address : 1 } );



which is more than 42 and upto 52..

5. Write a MongoDB query to arrange the name of the restaurants in ascending order along with all the columns.

```
db.restaurants.find().sort({name: 1});
```

6. Write a MongoDB query to arrange the name of the restaurants in descending order along with all the columns.

```
db.restaurants.find().sort({name: -1});
```

7. Write a MongoDB query to arranged the name of the cuisine in ascending order and for that same cuisine borough should be in descending order.

```
db.restaurants.find().sort({cuisine: 1, borough: -1});
```

8. Write a MongoDB query to know whether all the addresses contains the street or not.

```
db.restaurants.find({"address.street": {exists: true}});
```

9. Write a MongoDB query which will select all documents in the restaurants collection where the coord field value is Double.

```
db.restaurants.find({"address.coord": {"type": "double"}});
```

10. Write a MongoDB query which will select the restaurant id, name and grades for those restaurants which returns 0 as a remainder after dividing the score by 7.

```
db.restaurants.find({"grades.score": {"mod": [7, 0]}});
```

11. Write a MongoDB query to find the restaurant name, borough, longitude and attitude and cuisine for those restaurants which contains 'mon' as three letters somewhere in its name.

```
db.restaurants.find({name: {"regex": "mon"}}, {name: 1, borough: 1, "address.coord": 1, cuisine: 1});
```

12. Write a MongoDB query to find the restaurant name, borough, longitude and latitude and cuisine for those restaurants which contain 'Mad' as first three letters of its name.

```
db.restaurants.find({name: {"regex": "Mad"}}, {name: 1, borough: 1, "address.coord": 1, "address.lat": 1, cuisine: 1});
```



13. Write a MongoDB query to find the restaurants that have at least one grade with a score of less than 5.

db.restaurants.find({ "grades.score": { \$lt: 5 } });

14. Write a MongoDB query to find the restaurants that have at least one grade with a score of less than 5 and that are located in the borough of Manhattan.

db.restaurants.find({ "grades.score": { \$lt: 5 }, borough: "Manhattan" });

15. Write a MongoDB query to find the restaurants that have at least one grade with a score of less than 5 and that are located in the borough of Manhattan or Brooklyn.

db.restaurants.find({ "grades.score": { \$lt: 5 }, borough: { \$in: [ "Manhattan", "Brooklyn" ] } });

16. Write a MongoDB query to find the restaurants that have at least one grade with a score of less than 5 and that are located in the borough of Manhattan or Brooklyn, and their cuisine is not American.

db.restaurants.find({ "grades.score": { \$lt: 5 }, borough: { \$in: [ "Manhattan", "Brooklyn" ] }, cuisine: { \$ne: "American" } });

17. Write a MongoDB query to find the restaurants that have at least one grade with a score of less than 5 and that are located in the borough of Manhattan or Brooklyn, and their cuisine is not American or Chinese.

db.restaurants.find({ "grades.score": { \$lt: 5 }, borough: { \$in: [ "Manhattan", "Brooklyn" ] }, cuisine: { \$nin: [ "American", "Chinese" ] } });

18. Write a MongoDB query to find the restaurants that have a grade with a score of 2 and a grade with a score of 6.

db.restaurants.find({ "grades.score": { \$all: [ 2, 6 ] } });

19. Write a MongoDB query to find the restaurants that have a grade with a score of 2 and a grade with a score of 6 and are located in the borough of Manhattan.

db.restaurants.find({ borough: "Manhattan", "grades.score": { \$all: [ 2, 6 ] } });

20. Write a MongoDB query to find the restaurants that have a grade with a score of 2 and a grade with a score of 6 and are located in the borough of Manhattan or Brooklyn.

db.restaurants.find({ borough: { \$in: [ "Manhattan", "Brooklyn" ] }, "grades.score": { \$all: [ 2, 6 ] } });



21. Write a MongoDB query to find the restaurants that have a grade with a score of 2 and a grade with a score of 6 and are located in the borough of Manhattan or Brooklyn, and their cuisine is not American.

`db.restaurants.find({borough: {'$in': ['Manhattan', 'Brooklyn']}},  
cuisine: {'$ne': 'American'},  
'grades.score': {'$in': [2, 6]}}`

22. Write a MongoDB query to find the restaurants that have a grade with a score of 2 and a grade with a score of 6 and are located in the borough of Manhattan or Brooklyn, and their cuisine is not American or Chinese.

`db.restaurants.find({borough: {'$in': ['Manhattan', 'Brooklyn']}},  
cuisine: {'$nin': ['American', 'Chinese']},  
'grades.score': {'$in': [2, 6]}}`

23. Write a MongoDB query to find the restaurants that have a grade with a score of 2 or a grade with a score of 6.

`db.restaurants.find({'grades.score': {'$in': [2, 6]}})`

#### Sample document of 'movies' collection

```
{
  _id: ObjectId("573a1390f29313caabcd42e8"),
  plot: 'A group of bandits stage a brazen train hold-up, only to find a determined posse hot on their heels.',
  genres: [ 'Short', 'Western' ],
  runtime: 11,
  cast: [
    'A.C. Abadie',
    "Gilbert M. 'Broncho Billy' Anderson",
    'George Barnes',
    'Justus D. Barnes'
  ],
  poster: 'https://m.media-
amazon.com/images/M/MV5BMTU3NjE5NzYtYTYyNS00MDVmlWlWYjgtMmYwYWlxdDYyNzU2XkEyXkFqcG
deQXVyNzQzNzQxNzI@._V1_SY1000_SX677_AL_.jpg',
  title: 'The Great Train Robbery',
  fullplot: "Among the earliest existing films in American cinema - notable as the first film that presented a
narrative story to tell - it depicts a group of cowboy outlaws who hold up a train and rob the passengers.
They are then pursued by a Sheriff's posse. Several scenes have color included - all hand tinted."
```

| Evaluation Procedure  | Marks awarded |
|-----------------------|---------------|
| PL/SQL Procedure(5)   | 15            |
| Program/Execution (5) | 5             |
| Viva(5)               | 5             |
| Total (15)            | 15            |
| Faculty Signature     | P. S. M.      |